



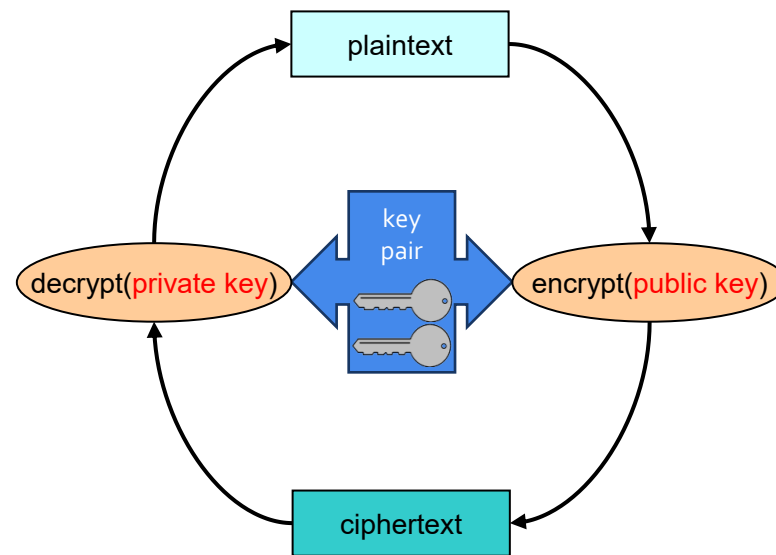
Asymmetric cryptography

Asymmetric (Block) Ciphers

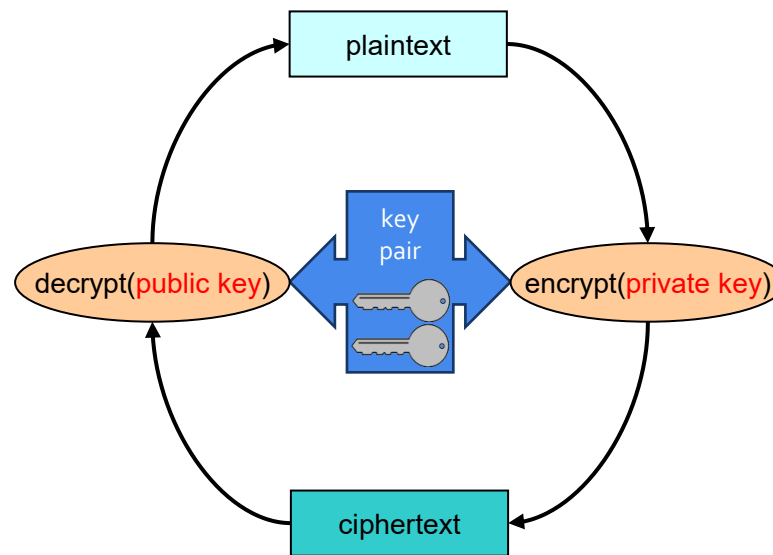
- Use key pairs
 - One private key (personal, not transmittable)
 - One public key, available to all
- Allow
 - Confidentiality without any previous exchange of secrets
 - Authentication
 - Of contents (data integrity)
 - Of origin (source authentication, or digital signature)

Operations of an asymmetric cipher

- Confidentiality

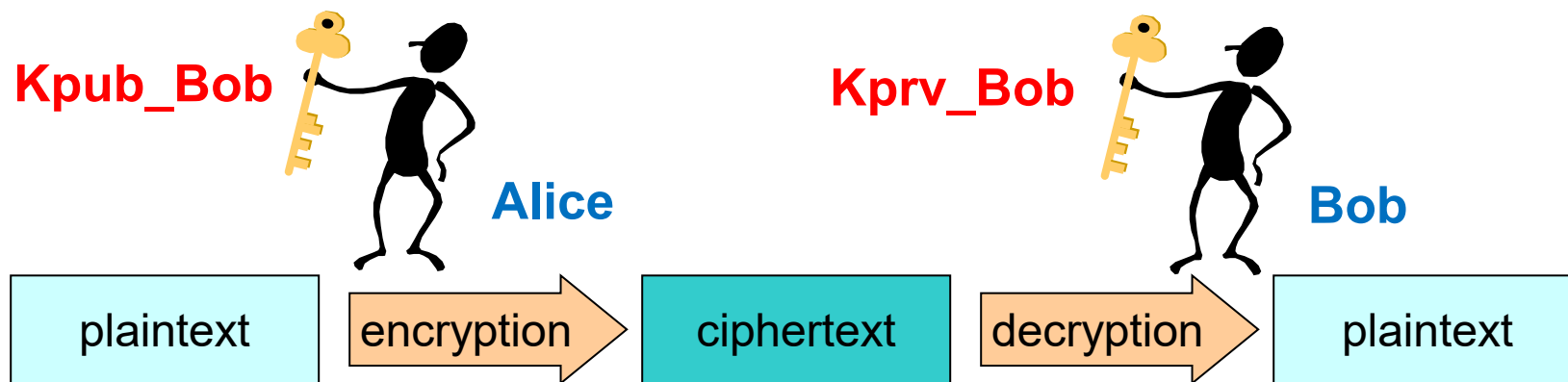


- Authentication (signature)



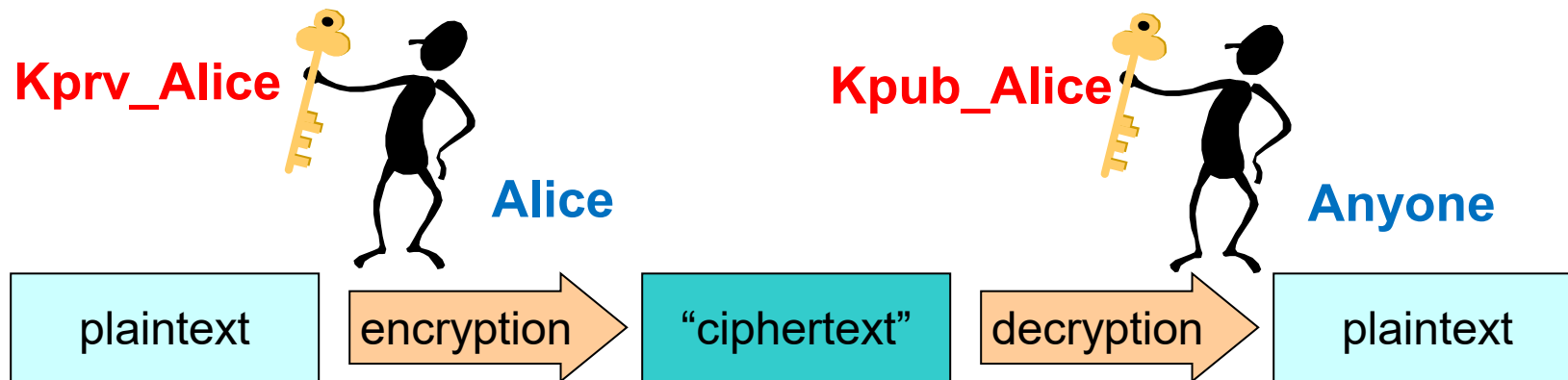
Use cases: secure communication

- Secure communication with a target (Bob)
 - Alice encrypts plaintext **P** with Bob's public key **Kpub_Bob**
Alice: $C = \{P\}_{K_{pub_Bob}}$
 - Bob decrypts ciphertext **C** with his private key **Kpriv_Bob**
Bob: $P' = \{C\}_{K_{priv_Bob}}$
 - **P'** should be equal to **P** (requires checking)
 - **Kpub_Bob** needs to be known by Alice



Use cases: signature

- Data signature by Alice
 - Alice encrypts plaintext **P** with her private key **K_{prv_Alice}**
Alice: $C = \{P\}_{K_{prv_Alice}}$
 - Anyone can decrypt cyphertext **C** with Alice's public key **K_{pub_Alice}**
Anyone: $P' = \{C\}_{K_{pub_Bob}}$
 - If $P' = P$, then **C** is **Alice's signature** of **P**
 - **K_{pub_Alice}** needs to be known by signature verifiers



Asymmetric ciphers

- Advantages
 - They are a fundamental authentication mechanism
 - They allow to explore features that are not possible with symmetric ciphers
- Disadvantages
 - Performance
 - Usually are very inefficient and memory consuming
- Problems
 - Trustworthy distribution of public keys
 - Lifetime of key pairs

Asymmetric ciphers

- Approaches: complex mathematic problems
 - Discrete logarithms of large numbers
 - Integer factorization of large numbers
- Most common algorithms
 - RSA
 - ElGamal
 - Elliptic curves (ECC)
- Other techniques with asymmetric key pairs
 - Diffie-Hellman (key agreement)

RSA (Rivest, Shamir, Adelman, 1978)

- Keys
 - Private: (d, n)
 - Public: (e, n)
- Public key encryption (confidentiality)
 - $C = P^e \bmod n$
 - $P = C^d \bmod n$
- Private key encryption (signature)
 - $C = P^d \bmod n$
 - $P = C^e \bmod n$

P, C are big numbers!

$$0 \leq P, C < n$$

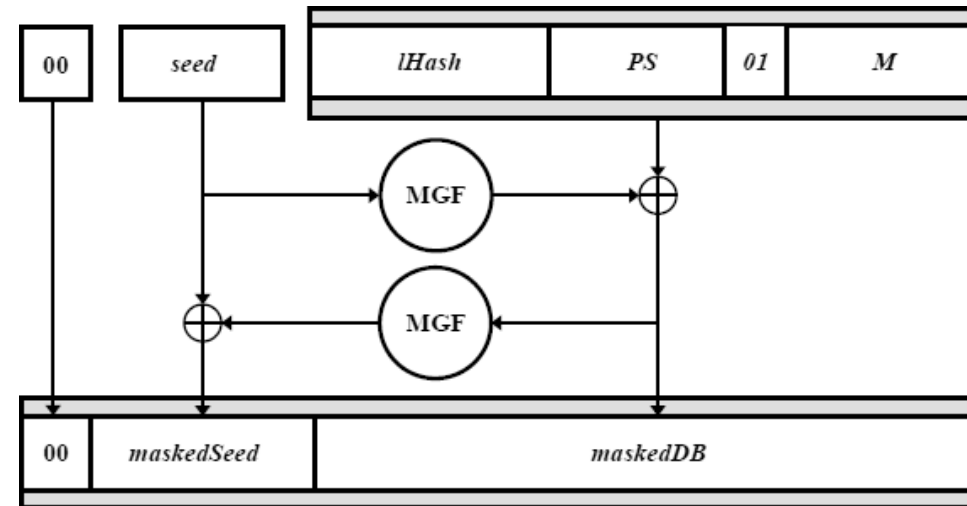
Hybrid encryption

- Combines symmetric with asymmetric cryptography
 - Use the best of both worlds, while avoiding problems
 - Asymmetric cipher: Uses public keys (but it is slow)
 - Symmetric cipher: Fast (but with weak key exchange methods)
- Method:
 - Obtain K_{pub} from the receiver
 - Generate a random K_{sym}
 - Calculate $C_1 = E_{sym}(K_{sym}, P)$
 - Calculate $C_2 = E_{asym}(K_{pub}, K_{sym})$
 - Send C_1 and C_2
 - C_1 = Text encrypted with symmetric key
 - C_2 = Symmetric key encrypted with the receiver public key
 - May also contain the IV

Randomization of asymmetric encryptions

- Non-deterministic (unpredictable) result of asymmetric encryptions
 - N encryptions of the **same value**, with the **same key**, should yield N **different results**
 - Goal:
 - Prevent the trial & error discovery of values encrypted w/ public keys
 - Prevent the cryptanalysis of the private key (noise)
- Approaches
 - Concatenation of the value to encrypt with two other values:
 - A fixed one (for integrity control)
 - A random one (for randomization)

Randomization of public key encryptions: OAEP (Optimal Asymmetric Encryption Padding)



- lHash: digest over Label
- seed: random
- PS: zeros
- M: plaintext
- MGF: Mask Generation Function
 - Similar to Hash, but with variable size

Diffie-Hellman key agreement (1976)



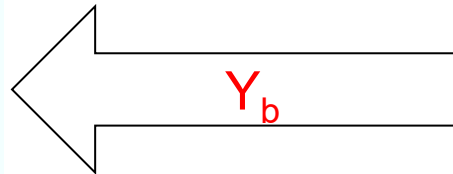
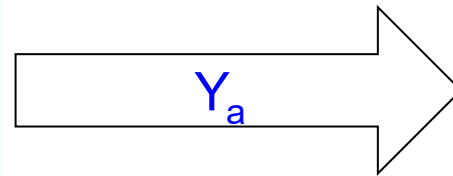
q (large prime)
 α (primitive root mod q)



a = random

$$Y_a = \alpha^a \text{ mod } q$$

$$K_{ab} = Y_b^a \text{ mod } q$$



$$K_{ab} = K_{ba}$$

b = random

$$Y_b = \alpha^b \text{ mod } q$$

$$K_{ba} = Y_a^b \text{ mod } q$$

DH key agreement: MitM attack



a = random

$$Y_a = \alpha^a \bmod q$$

$$K_{ac} = Y_c^a \bmod q$$



c = random

$$Y_c = \alpha^c \bmod q$$

$$K_{ca} = Y_a^c \bmod q$$

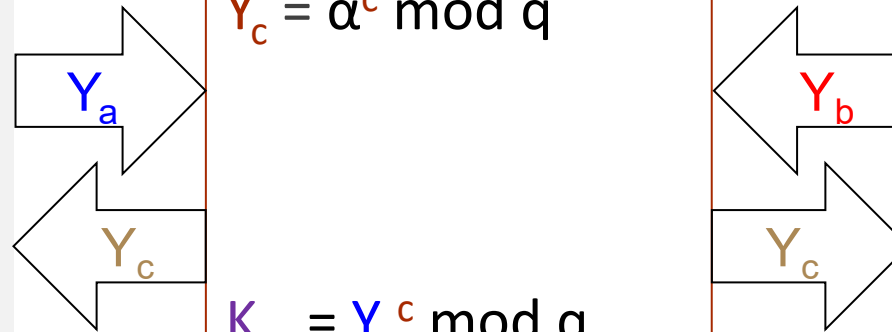
$$K_{cb} = Y_b^c \bmod q$$



b = random

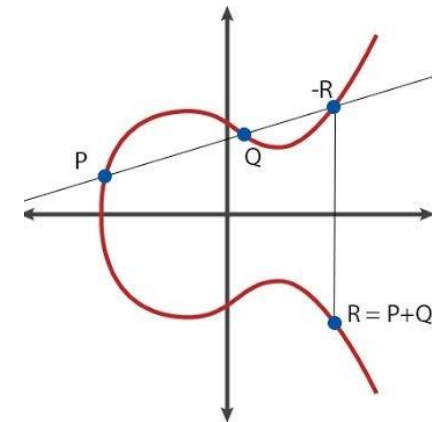
$$Y_b = \alpha^b \bmod q$$

$$K_{bc} = Y_c^b \bmod q$$



Elliptic Curve Cryptography (ECC)

- Elliptic curves are specific functions
 - They have a generator (G)
 - A private key K_{prv} is an **integer** with a maximum of bits allowed by the curve
 - A public key K_{pub} is a **point** $(x,y) = K_{\text{prv}} \times G$
 - Given K_{pub} , it should be hard to guess K_{prv}
- Curves
 - NIST curves (15)
 - P-192, P-224, P-256, P-384, P-521
 - B-163, B-233, B-283, B-409, B-571
 - K-163, K-233, K-283, K-409, K-571
 - Other curves
 - Curve25519 (256 bits)
 - Curve448 (448 bits)



ECDH: DH with ECC



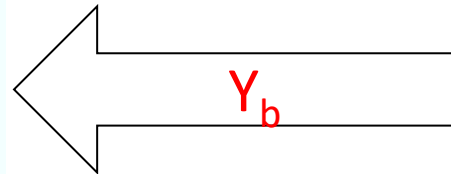
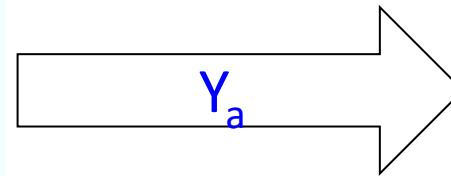
ECC curve $\rightarrow G$



a = random

$$Y_a = aG$$

$$K_{ab} = aY_b = abG$$



$$K_{ab} = K_{ba}$$

b = random

$$Y_b = bG$$

$$K_{ba} = bY_a = baG$$

ECC public key encryption

- Combines hybrid encryption with ECDH
- Method:
 - Obtain $K_{\text{pub_recv}}$ from the receiver
 - Generate a random $K_{\text{prv_send}}$ and the corresponding $K_{\text{pub_send}}$
 - Calculate $K_{\text{sym}} = K_{\text{prv_send}} K_{\text{pub_recv}}$
 - $C = E(P, K_{\text{sym}})$
 - Send C and $K_{\text{pub_send}}$
 - Receiver calculates $K_{\text{sym}} = K_{\text{pub_send}} K_{\text{prv_recv}}$
 - $P = D(C, K_{\text{sym}})$

Bibliografia

Leitura recomendada:

- Segurança em Redes Informáticas, A. Zúquete: Chap. 2

Leitura adicional:

- [Diffie-Hellman key exchange](#)
- [Rivest-Shamir-Adleman \(RSA\) cryptosystem](#)
- [RSA Factoring Challenge](#)
- [PKCS #1](#)
- [Optimal asymmetric encryption padding](#)
- [Digital signature](#)
- [Elliptic-curve Diffie–Hellman](#)
- [Elliptic curve](#)
- [Elliptic curve point multiplication](#)
- [Elliptic curve cryptography](#)